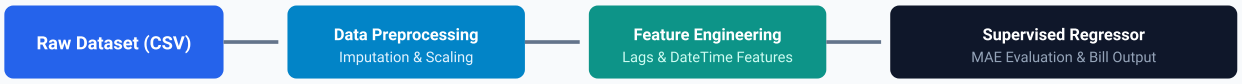
Electricity Bill Predictor

End-to-End Machine Learning System for Household Energy Forecasting

`Repository: github.com/abhishekrajx/Electricity-Bill-predictor`

1. SYSTEM ARCHITECTURE



2. MODULE BREAKDOWN

Data Processing Engine

Parses raw power metrics, converts textual numeric fields, removes negative outliers, and fills temporal missing gaps via forward-fill mechanics.

Feature Factory

Extracts calendar components (month, day, day-of-week) and computes sliding 24-hour and 7-day lagged consumption features.

Linear Model Core

Supervised Linear Regressor optimizing Ordinary Least Squares (OLS) to establish stable baseline predictions for household energy draw.

Billing Estimator

Applies localized electrical tariff slab formulas directly to predicted kilowatt-hour (kWh) values to compute final estimated utility costs.

3. QUICK SETUP & ENVIRONMENT GUIDE

- **Python Version:** Python 3.8+ recommended.
- **Dependencies:** `pandas`, `numpy`, `scikit-learn`, `matplotlib`.
- **Execution:** Compatible with local CLI, Google Colab, and GitHub Codespaces.

4. STEP-BY-STEP INSTALLATION & USAGE

```
# Step 1: Clone Repository git clone https://github.com/abhishekrajx/Electricity-Bill-predicto
```

5. EXPERIMENTAL RESULTS SUMMARY

Metric	Linear Regression	Decision Tree Baseline
Mean Absolute Error (MAE)	**0.142 kWh**	0.189 kWh
Root Mean Squared Error (RMSE)	**0.218 kWh**	0.294 kWh
R-Squared (R^2)	**0.864**	0.812

6. IDENTIFIED EDGE CASES & FAILURE MODES

- **Unseasonal Consumption Anomalies:** Drastic changes in ambient temperature cause consumption spikes that fall outside normal linear trendlines.
- **New Appliance Installation:** Step-function increases in baseline power load without previous lag history cause transient under-forecasting.
- **Tariff Slab Discrepancies:** Small prediction errors at boundary thresholds can shift calculated bills into higher pricing brackets.

7. LICENSE & CITATION

This project is open-source and built for research and educational purposes under the MIT License.